(FP1) 2. Rational functions and graphs 专题作业

NO. 1

Let a be a positive constant.

a. The curve C_1 has equation $y = \dfrac{x-a}{x-2a}$. [2]
Sketch C_1.

The curve C_2 has equation $y = \left(\dfrac{x-a}{x-2a}\right)^2$. The curve C_3 has equation $y = \left|\dfrac{x-a}{x-2a}\right|$.

b. i. Find the coordinates of any stationary points of C_2. [3]

ii. Find also the coordinates of any points of intersection of C_2 and C_3 . [3]

c. Sketch C_2 and C_3 on a single diagram, clearly identifying each curve. Hence find the set of values of x for which $\left(\dfrac{x-a}{x-2a}\right)^2 \leqslant \left|\dfrac{x-a}{x-2a}\right|$. [5]

NO. 2

The curve C has equation $y = \frac{x^2 + x - 1}{x - 1}$.

a. Find the equations of the asymptotes of C. [3]

b. Show that there is no point on C for which 1 < y < 5. [4]

c. Find the coordinates of the intersections of C with the axes, and sketch C. [3]

d. Sketch the curve with equation $y = \left| \frac{x^2 + x - 1}{x - 1} \right|$. [2]

NO. 3

Let a be a positive constant.

a. Sketch the curve with equation $y = \frac{ax}{x+7}$. [2]

b. Sketch the curve with equation $y = \left|\frac{ax}{x+7}\right|$ and find the set of values of x for which $\left|\frac{ax}{x+7}\right| > \frac{a}{2}$. [4]

NO. 4

The curve C has equation

$$y = \frac{x^2 + ax - 1}{x + 1},$$

where a is constant and $a > 1$.

i. Find the equations of the asymptotes of C. [3]

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ii. Show that C intersects the x-axis twice. [1]

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iii. Justifying your answer, find the number of stationary points on C. [2]

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iv. Sketch C, stating the coordinates of its point of intersection with the y-axis. [3]

NO. 5

A curve C has equation $y = \frac{ax^2 + x - 1}{x - 1}$, where a is a positive constant.

a. Find the equations of the asymptotes of C. [3]

b. Show that there is no point on C for which $1 < y < 1 + 4a$. [4]

c. Sketch C. You do not need to find the coordinates of the intersections with the axes. [3]

NO. 6

The curve C has equation $y = \frac{x^2}{x-3}$.

a. Find the equations of the asymptotes of C. [3]

b. Show that there is no point on C for which 0 < y < 12. [4]

c. Sketch C. [2]

d. i. Sketch the graphs of $y = \left|\frac{x^2}{x-3}\right|$ and $y = |x| - 3$ on a single diagram, stating the coordinates of the intersections with the axes. [4]

ii. Use your sketch to find the set of values of c for which $\left|\frac{x^2}{x-3}\right| \leqslant |x| + c$ has no solution. [1]

NO. 7

The curve C has equation $y = \dfrac{4x+5}{4-4x^2}$.

a. Find the equations of the asymptotes of C. [2]

b. Find the coordinates of any stationary points on C. [4]

c. Sketch C, stating the coordinates of the intersections with the axes. [3]

d. Sketch the curve with equation $y = \left|\dfrac{4x+5}{4-4x^2}\right|$ and find in exact form the set of values of x for which $4|4x+5| > 5|4-4x^2|$. [6]